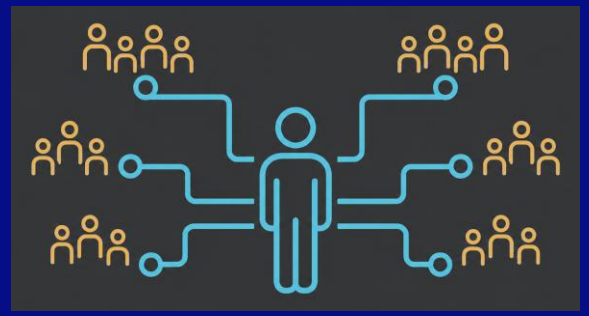
University of Otago
ŌTĀKOU WHAKAIHU WAKA
Te Kura Rorohiko
School of Computing
Te Reo Inarahi me te Mātai Wetereo
English and Linguistics



***DHUM101***

***Socially Responsible Computing***

# Lecture 3
# Argumentation

- Hunter Hatfield
Semester 2, 2026

# Today

Story 1: What We Believe Matters

Story 2: Reason is what makes humans special and we're horrible at it

Story 3: Human Reason and Its Function

Story 4: Assessing Arguments: Truth and Rhetoric

# Story 1: What We Believe Matters

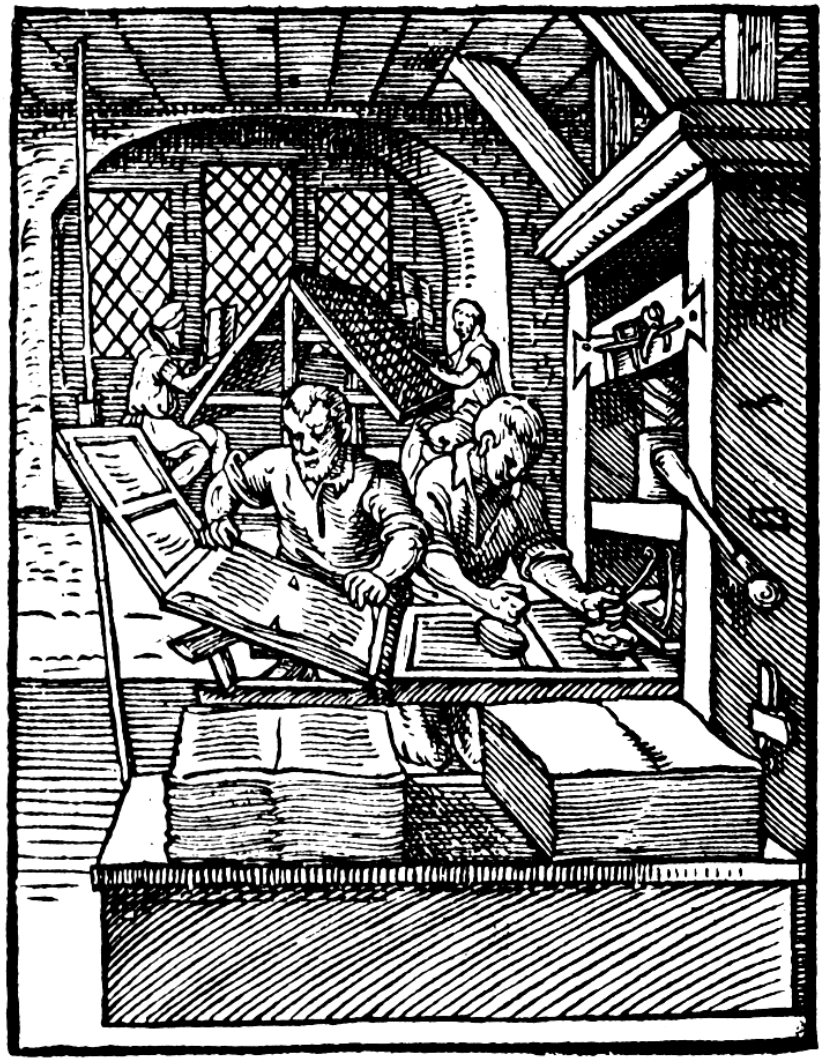
# What We Believe Matters

The Printing Press. (mid 1400s)

Scientific revolution. Allowed distribution of ideas

Political Movements. More people could ‘speak’

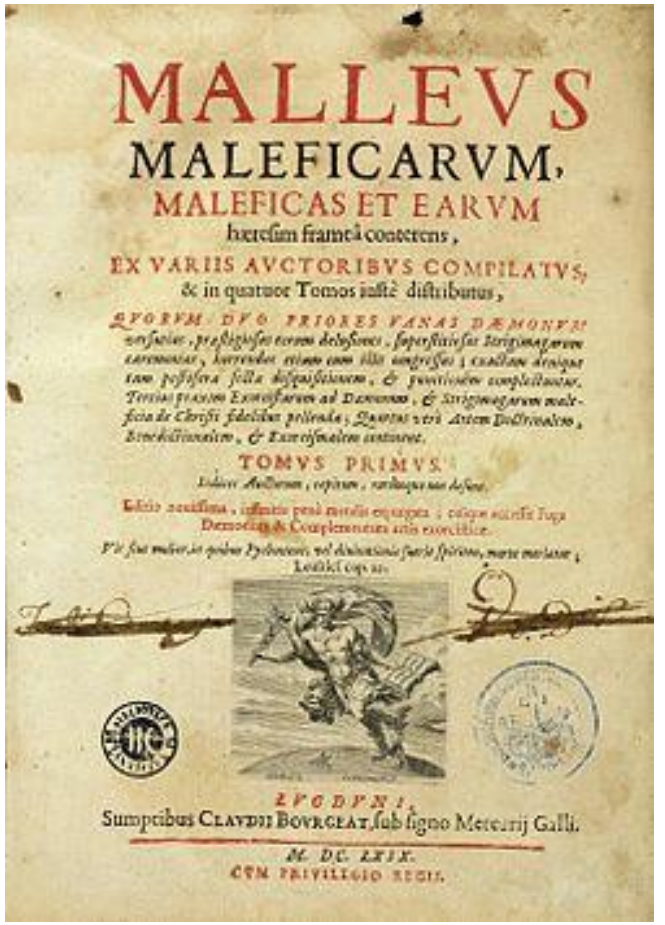
Nationalism. Standard languages, standard nations.



# What We Believe Matters

The Hammer of Witches. Published just decades later.

Published 100 kms south of Gutenberg




https://commons.wikimedia.org/w/index.php?curid=35858408

# What We Believe Matters

**The Hammer of Witches.** Published just decades later.

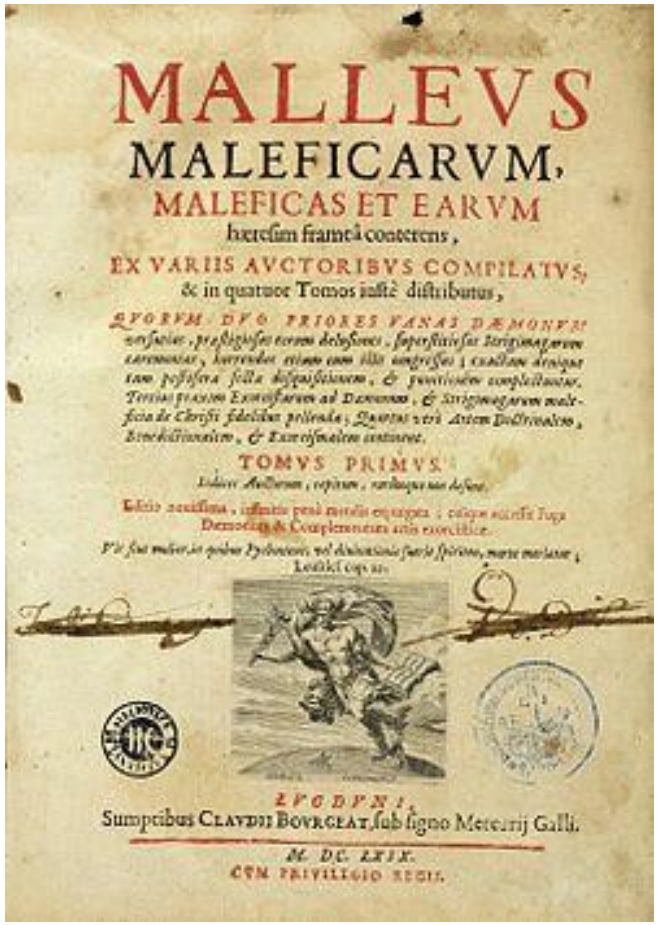
Published 100 kms south of Gutenberg

**Working with demons.** Worked with evil supernatural

**Horrible crimes.** These people would kill babies and more

**Mostly Women.** Women had weak wills and were lustful

**Only a Matter of Time.** Most women would fall eventually.




https://commons.wikimedia.org/w/index.php?curid=35858408

# What We Believe Matters

**The Hammer of Witches.** Published just decades later.

Published 100 kms south of Gutenberg

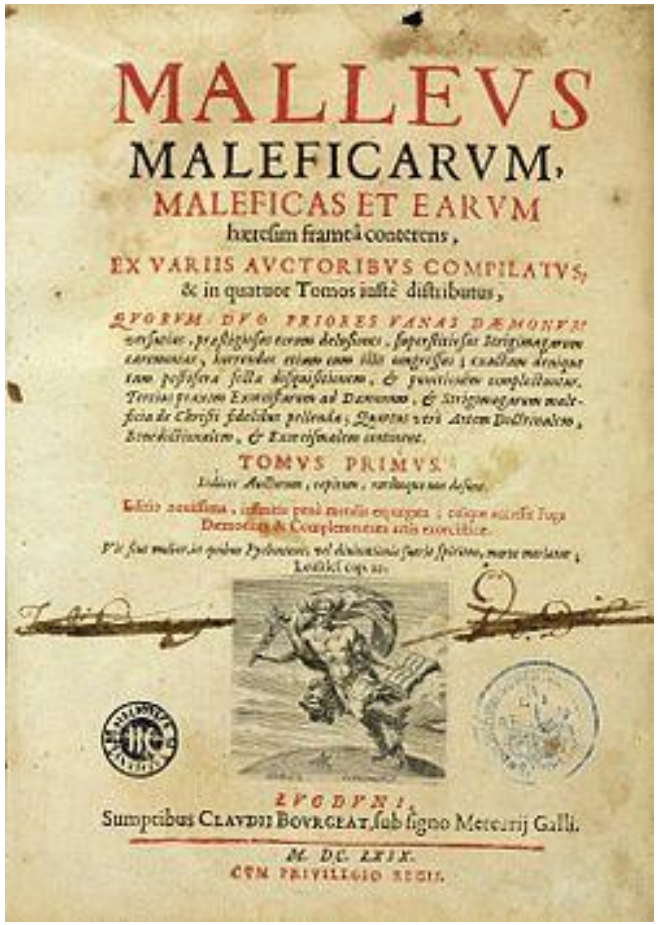
**Working with demons.** Worked with evil supernatural

**Horrible crimes.** These people would kill babies and more

**Mostly Women.** Women had weak wills and were lustful

**Only a Matter of Time.** Most women would fall eventually.

Tens of thousands (3/4 women) were killed




https://commons.wikimedia.org/w/index.php?curid=35858408

# Beliefs Affect Actions

Serious choices Medical choices, economic choices, safety choices, etc.

Claims People Make

- ‘AI uses too much energy and is bad for the environment.’
- ‘AI will bring a world of great abundance.’
- ‘AI will replace 50% of workers in the next 15 years.’
- ‘AI is not living up to expectations and businesses are rehiring the workers they fired.’

Annoyances You think a Python function works one way, while it really works another way, and you waste hours on an assignment.

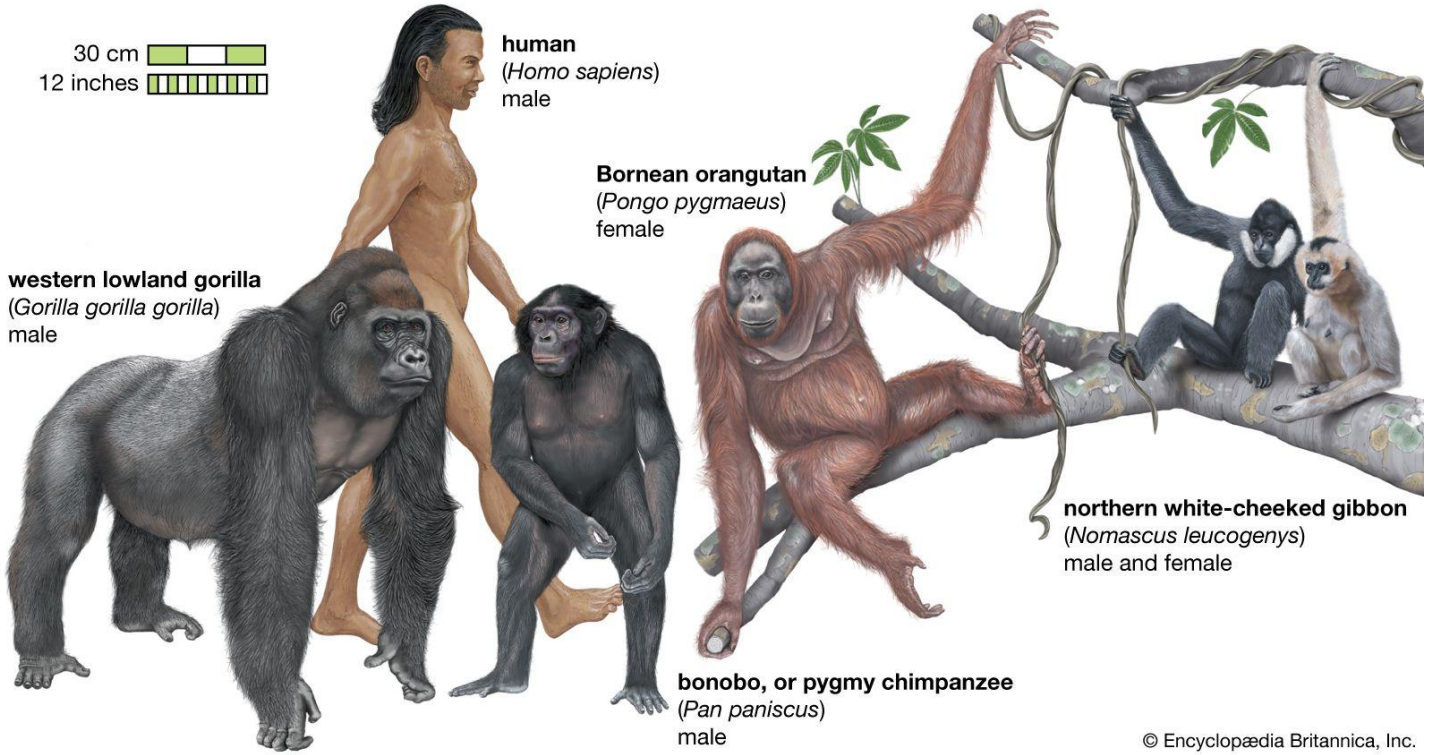
# Basic to Human Conversation

- Conversation only works when certain patterns are followed
- What you say is relevant to what the other person said
- What you say is understandable
- What you say is true.

Responsibility Whenever you speak, you are responsible for what you say. If someone concludes you are never responsible, they stop speaking to you.

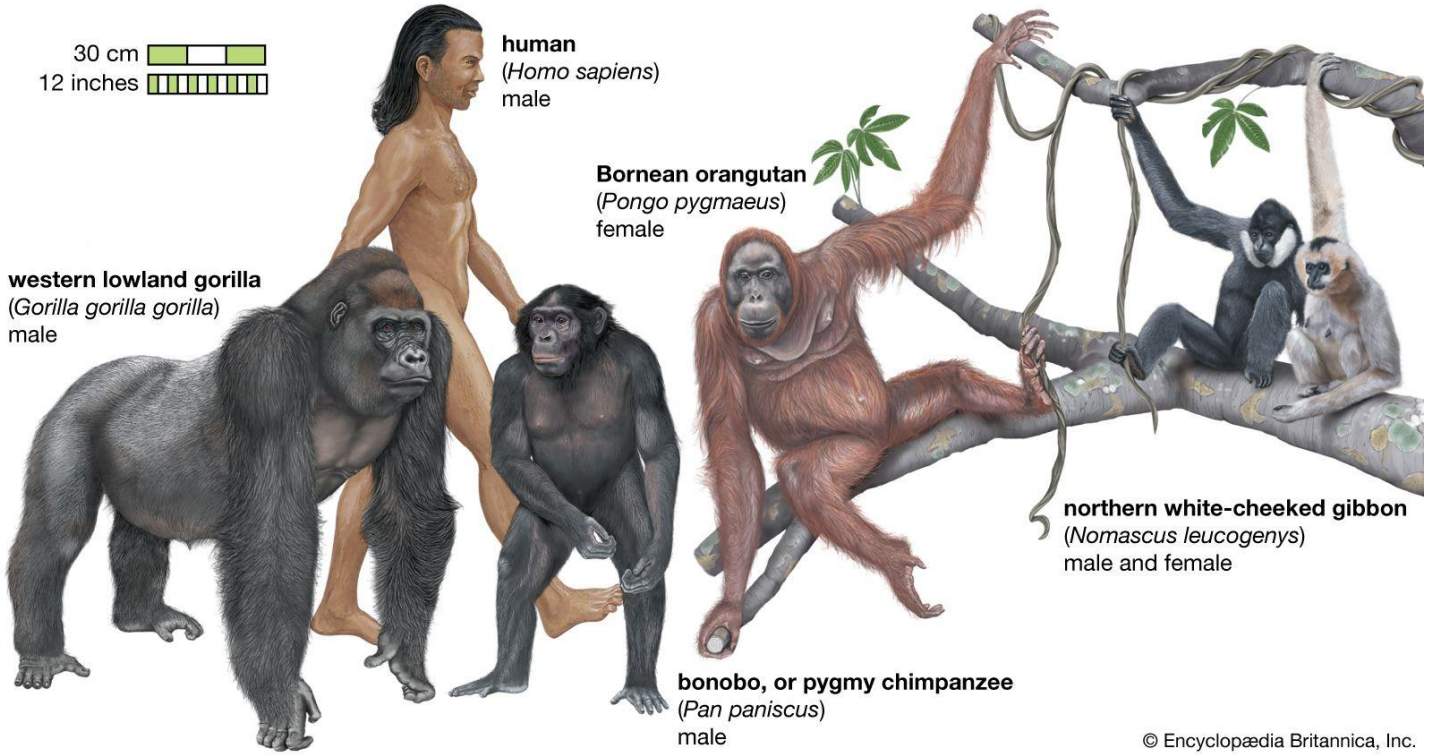
# Story 2: Reason is what makes humans special and we’re horrible at it

# What Makes Humans Different?




https://www.britannica.com/animal/primate-mammal

# What Makes Humans Different?




## Reason:

- Allows us to build houses and do farming and launch rockets.
- Reason allows us to figure out true from false.

https://www.britannica.com/animal/primate-mammal

# Reason Makes Us Unique but We're BAD At It

Logic Developed by peoples around the world for over 2000 years

Formal logic

- If you combine a particular form (the grammar of logic) with
- True premises
- You get a True conclusion

Informal Logic "The study of reasoning and inference as they occur in public discussion and debate, educational and intellectual pursuits" (https://plato.stanford.edu/entries/logic-informal/)

# Reason Makes Us Unique but We're BAD At It

Logic Developed by peoples around the world for over 2000 years

Formal logic

- If yo
- Tru
- You

BAD!!!

Informal Logic "The study of reasoning and inference as they occur in public discussion and debate, educational and intellectual pursuits" (https://pl

BAD!!!

# Formal Logic

Logic The form of an argument is important.

Example 1

- If something is a square, it is a rectangle
- X is a square
- X is a rectangle

Example 2

- If something is a square, it is a circle
- X is a square
- X is a circle

# Formal Logic

Logic The form of an argument is important.

## Example 1

- If something is a square, it is a rectangle
- X is a square
- X is a rectangle

Valid

Sound

## Example 2

- If something is a square, it is a circle
- X is a square
- X is a circle

Valid

Unsound

# Formal Logic

Logic The form of an argument is important.

## Example 1

- If something is a square, it is a rectangle
- X is a square
- X is a rectangle

Valid

Sound

## Example 2

- If something is a square, it is a circle
- X is a square
- X is a circle

Valid

Unsound

If the logical form is valid AND the premises are true, then the conclusion will be true

# Affirming the Consequent

Logic A common formal logic error is affirming the consequent

## Example 1

- Plants produce oxygen as a product of photosynthesis.
- We've found a planet with oxygen,
- Therefore, it has plants

## Example 2

- If a person is hungry, they will go to the grocery story.
- A person is at the grocery store,
- Therefore, they were hungry.

# Affirming the Consequent

Logic A common formal logic error is affirming the consequent

## Example 3

- If my alarm clock goes off, I will wake up.
- I am awake.
- Therefore, my alarm clock went off.

## Example 4

- If X causes Y, then X correlates with Y.
- X correlates with Y
- X causes Y

# Correlation Does Not Equal Causation

True If one event causes another event, they will be correlated

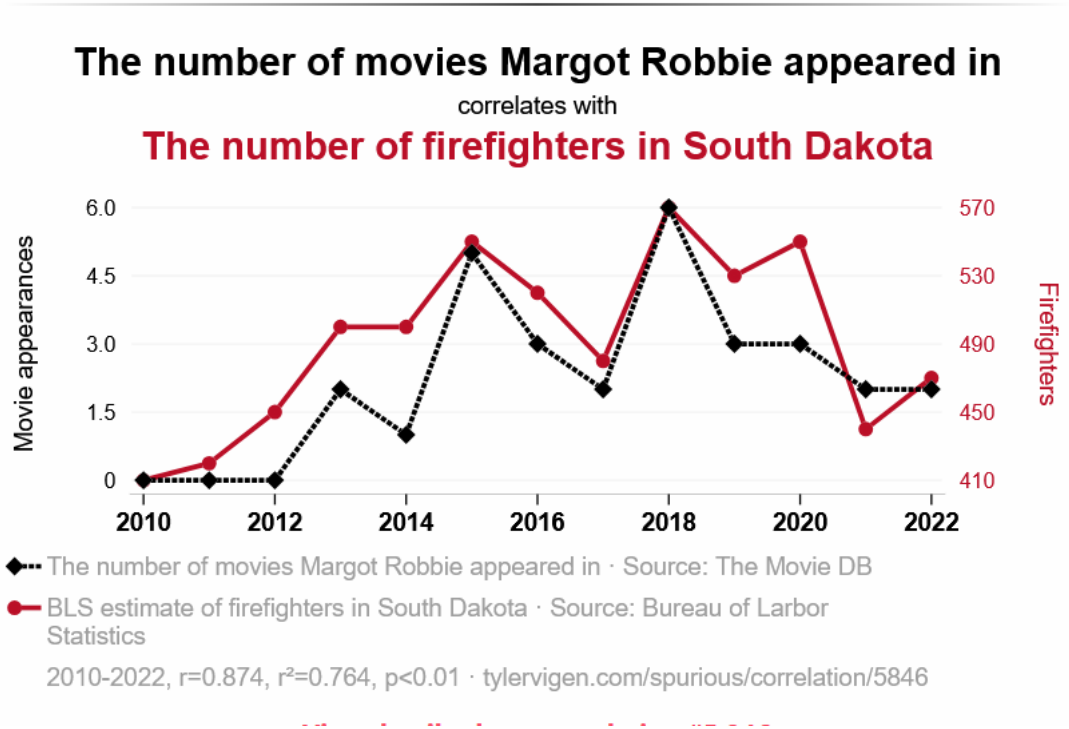
Maybe True If one event correlates with another event, that event caused the other event

# Correlation Does Not Equal Causation

True If one event causes another event, they will be correlated

Maybe True If one event correlates with another event, that event caused the other event






Great Site!

- https://www.tylervigen.com/spurious-correlations
- https://www.tylervigen.com/spurious/variable?id=26637

https://www.biography.com/actors/margot-robbie

# Vague Premises

**Ambiguity** The most common problem is that it's hard to tell if a premise is true because the terms are vague

**'We must adopt AI or go obsolete.'**

- What do they mean by AI? Large language models? Any machine learning? AGI? What sort of LLMs? What sort of machine learning?

**The world will be better when AGI is achieved.**

- What is AGI? Is it when a single model can do all things a human can do? Many things? Which ones? Why does it need to be one model, and not many models? Better how? All of the world?

# Informal Logic

Informal Logic "The study of reasoning and inference as they occur in public discussion and debate, educational and intellectual pursuits"

This is where we usually 'go wrong'

- Confirmation Bias
- Strawman Fallacy
- False Dichotomy
- Appeals and Attacks

# Confirmation Bias

The tendency to look for evidence that what you believe is right




The Most Common Cognitive Bias

https://www.youtube.com/watch?v=vKA4w2O61Xo

# Confirmation Bias

The tendency to look for evidence that what you believe is right

- If you think swans are white, check that by looking for white swans.

## Example

- Imagine six pieces of evidence: three support claim X, 3 support claim not-X.
  - Two people come in who already believe X or not-X. Do they leave thinking that the evidence is mixed? Sometimes, but often people instead select the evidence that agrees with them and find reasons to disregard the evidence that disagrees with them.

# Strawman Fallacy

**Want to gain trust** – Show evidence for your argument AND against it. Then show how the against is wrong.

**Refuting the best is hard** – Create a version of the opponent that is weak. Refute that!

**You have refuted a Straw Man**

# False Dichotomy (Dilemma)

Dichotomy – two opposite things

False dichotomy – Present **two** opposite choices when there are in reality many more.

Yours is best by elimination –

- Only two choices.
- One is REALLY bad.
- As long as yours isn't THAT bad, then it's the best choice.

'Either we adopt AI or we become obsolete to those using AI.'

'Either we scan the faces of customers or our employees will be unsafe.'

# Appeals and Attacks

**Appeal to emotion** – Trying to make you angry, fearful, excited, etc., rather than depending on true premises and valid logical form

**Ad hominem** – Attacking the person giving the argument, rather than the argument.

- Person X believes Y. Person X is an idiot (a liar, a bad person, etc.), so Y is false.

**Appeal to authority**

- Person X believes Y. Person X is a powerful person, so Y is true.
  - Person X is an expert on this matter, so Y is true.

# Informal Logic

Informal Logic "The study of reasoning and inference as they occur in public discussion and debate, educational and intellectual pursuits"

This is where we usually 'go wrong'

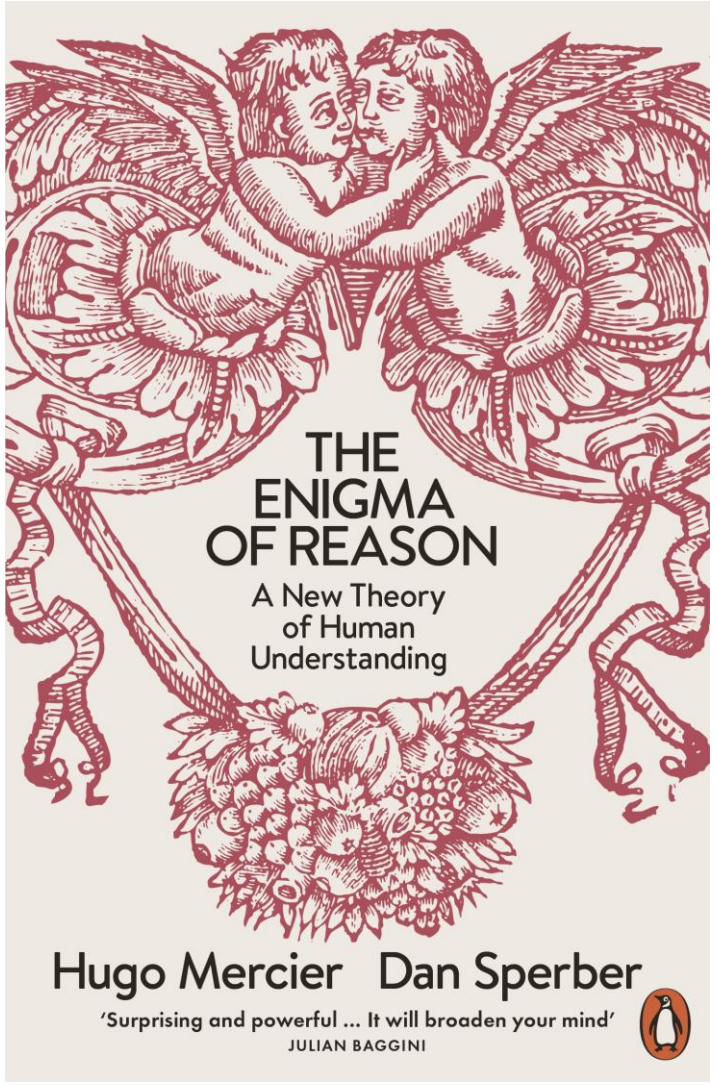
- Confirmation Bias
- Strawman Fallacy
- False Dichotomy
- Appeals and Attacks

- https://en.wikipedia.org/wiki/List_of_fallacies

# Story 3: Human Reason and Its Function

# Is Story 2 Wrong?

Why do humans reason at all?

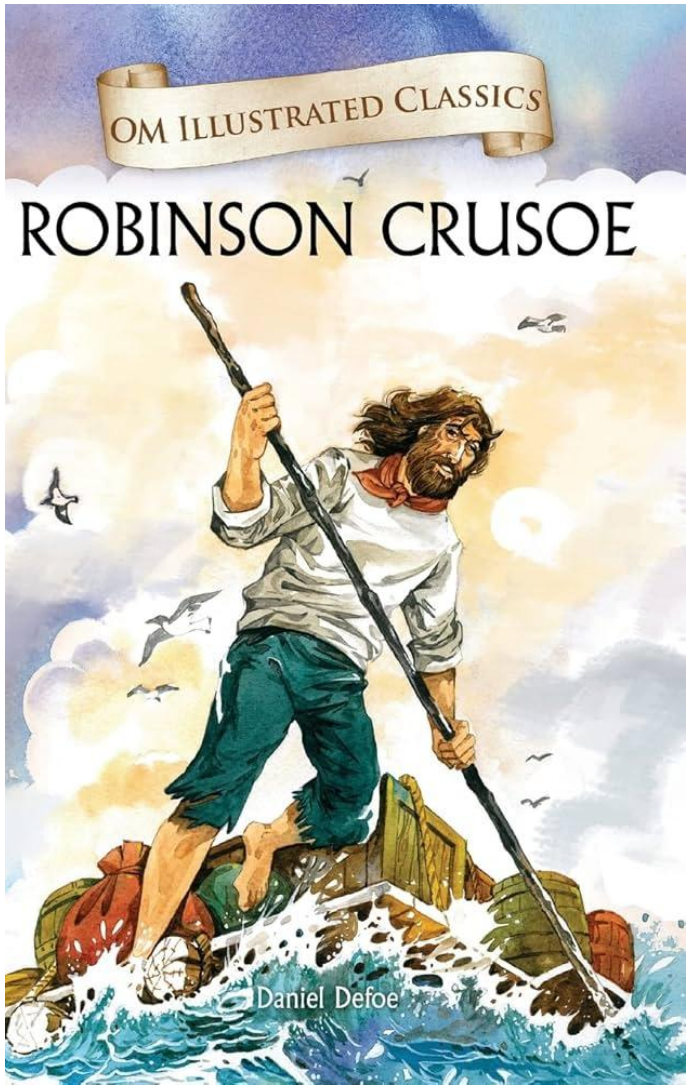


# Is Story 2 Wrong?

## Reason is for a person to figure out truth

‘In any natural history of the human species, language would stand out as the preeminent trait. To be sure, a solitary human is an impressive problem solver and engineer. But a race of Robinson Crusoes would not give an extraterrestrial observer all that much to remark on. What is truly arresting about our kind is better captured in the story of the Tower of Babel, in which humanity, speaking in a single language, came so close to reaching heaven that God himself felt threatened.’

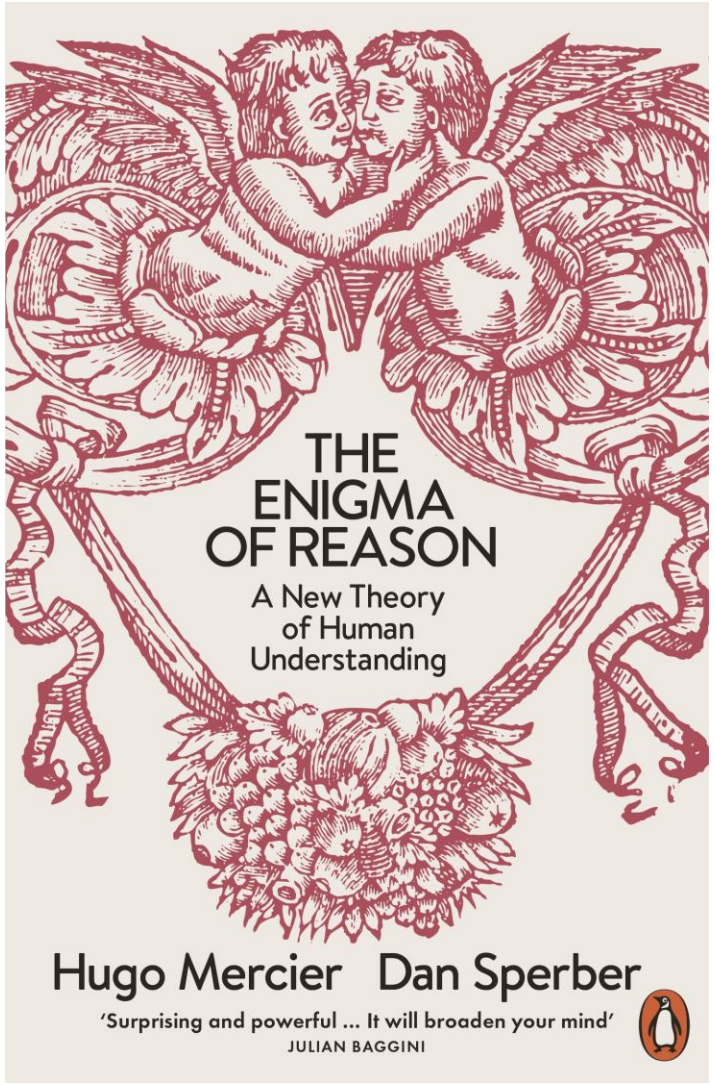
Cognitive scientist Steven Pinker, *The Language Instinct*, p. 16



# Reason is for Interaction

## Humans cooperate

- We can do more together than apart
- Fundamental to human activity
- Single people don't sail to the stars
- Groups of people do



# Reason is for Interaction

## How do we cooperate?

- We divide up tasks
- We have expectations for one another
- These tasks and expectations allow us to accomplish shared goals.
- Therefore
  - We need to communicate to others what we believe and why
  - We need to justify our actions to others
- But
  - We do not want to fall victim to others cheating, taking, not contributing
  - So we scrutinise them carefully

# Reason is for Interaction

## Confirmation bias

- Is a description of spending energy on others' beliefs, not on ours

## Appeal to authority

- We have to trust others as experts (or we're not cooperating at all)
- Who is trustworthy? Who is an expert? Whose ideas match what I think is true?
- We trust ourselves

## Appeal to emotion

- Reasoning is to get people to do things. Appeal to emotion, morality, consequences – we need people to care.

# Reason is for Interaction

## Alone we make mistakes

- If we are by ourselves, we don't notice our own errors

## In a community of open discussion

- Other people can point out our mistakes to us

## Results are better

- A group of people with differing beliefs talking opening usually results in better outcomes for the group

# Reason is for Interaction

## Alone we make mistakes

- If we are by ourselves, we don't notice our own errors

## In a community of open discussion

- Other people can point out our mistakes to us

## Results are better

- A group of people with differing beliefs talking opening usually results in better outcomes for the group

But cults!

But dictators!

Einstein!

# Story 4: Assessing Arguments: Truth and Rhetoric

# How should we assess something?

Assessing in Principle

Assessing in Practice

# How should we assess something?

In Principle – Assess everything fully to make sure every premise is true and there are no mistakes.

Ad Practice – This is impossible (or at least VERY challenging!)

- Premises have Premises, infinitely? Indefinitely
- You have to assume some things
- You will be better at assessing some claims than others
- Must rely on accepted knowledge AND experts
  - Expertise is specific

# How should we assess something?

Everyone has a bias – People state things for a purpose. That purpose affects what they say and how they say it

- Having a bias does not mean the claims are false, but it does help you find likely problems

Bias examples

  - If someone argues for investing in technology, do they own a technology company?
  - If someone critiques the environmental impact of a technology, do they have a background in environmental advocacy?
  - If someone worries about AI companies training on published books, do they work for a publisher?

# How should we assess something?

What do they want you to do? If a speaker has a purpose, what is that purpose? What actions do they want you to take?

- May be you directly, may be other listeners or readers
- Sometimes directly stated; sometimes implicit
- Inaction is an action
  - A lot of misinformation is not to convince you to do anything but simply to create enough doubt that you do not do anything

Study what the consequences are – If we do as they suggest, what are the results

- Does something good happen for the speaker? For the listener?
- Are some groups affected more than other groups?
- Do you know? Does anyone know?

# How should we assess something?

**Wanting action is okay** We do need to cooperate; we do need to act

- Sometimes you need to have fully cited references and explicit logic
- Sometimes you need to get people motivated
- Think of the purpose of the article

**Assemble the evidence yourself**

- Often people sprinkle evidence throughout
- To know the evidence for yourself, assemble it into an order

# How should we assess something?

Premises Evidence

- Evidence 1
- Evidence 2
- Evidence 3

Conclusion (claims)

- Therefore, this is true

# Assessment

# Structure Report Assessment

Due Date *Monday 10 August, 4:30pm, Submit in Aoroa*

What to do? Read one of four articles (one is a video)

- Sam Altman  the Gentle Singularity
  - https://blog.samaltman.com/the-gentle-singularity
- Cory Doctorow  Enshittification on the Daily Show (language notice)
  - https://www.youtube.com/watch?v=d2e-c9SF5nE
- David Krueger  No, AI is not Inevitable. We should stop it while we can.
  - https://www.usatoday.com/story/opinion/2026/01/24/ai-chip-manufacturing-data-centers-humanity/88215945007/
- Casey Fielder  AI has social consequences, but who pays the price?
  - https://theconversation.com/ai-has-social-consequences-but-who-pays-the-price-tech-companies-problem-with-ethical-debt-203375

Answer a set of questions Linked in Aoroa, practiced in tutorial. **Word count:** 2100.